# Bluestock Mutual Fund Analytics

## Capstone Project — Final Report - Muppuri Gopi

| Field | Value |
|---|---|
| Domain | Mutual Fund / Fintech Analytics |
| Tools | Python, Pandas, NumPy, SciPy, Matplotlib, Seaborn, SQLite, SQLAlchemy, Jupyter, Power |
| Database | bluestock_mf.db (SQLite, 7-table schema) |
| Dashboard | 4-page interactive Power BI dashboard |
| Date | July 2026 |

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# 1. Executive Summary:

This project analyzes mutual fund performance, NAV history, investor transactions, SIP continuity, and portfolio holdings using Python, SQL, SQLite, and Power BI. The objective is to generate actionable investment insights through descriptive analytics, risk metrics, and interactive dashboards.

The platform ingests data across 40 fund schemes from 10 fund houses, cleans and loads it into a normalized SQLite database, computes performance and risk metrics (CAGR, Sharpe, Sortino, Alpha, Beta, Maximum Drawdown, Historical VaR), and surfaces the results through a 4-page interactive Power BI dashboard with drill-through and KPI cards.

## Headline Numbers

| Metric | Value |
|---|---|
| Fund schemes tracked | 40 |
| Fund houses | 10 |
| NAV records | 41,720 |
| Investor transactions | 35,144 |
| Top fund scorecard | 98.1 / 100 (Axis Large Cap Fund) |
| Investors flagged at-risk for SIP lapse | 1,609 of 1,610 eligible |

## 2. Project Objectives:

- Build an end-to-end analytics pipeline from raw CSVs to a queryable SQLite database.
- Clean and transform financial datasets: NAV history, transactions, performance, portfolio holdings.
- Perform exploratory data analysis on NAV, AUM, SIP inflow, and investor behaviour.
- Calculate advanced risk metrics including Historical VaR, Rolling Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Sector HHI concentration.
- Build an interactive Power BI dashboard with KPI cards, slicers, and drill-through pages.
- Produce a simple, rule-based fund recommendation model.

## 3.Problem Statement:

Retail investors and advisors face several recurring problems when trying to make data-driven mutual fund decisions:

| # | Problem | This Project's Response |
|---|---|---|
| P1 | NAV, AUM, and SIP data are fragmented across separate files/APIs with no unified database. | A single ETL pipeline consolidates all sources into one SQLite database. |
| P2 | Investors cannot easily compare funds on a risk-adjusted basis without manual computation. | A computed Fund Scorecard blends CAGR, Sharpe, Alpha, expense ratio, and drawdown into one score. |
| P3 | Most investors cannot tell whether their fund is actually beating its benchmark index. | Alpha/Beta regression against NIFTY 100 TRI and a benchmark comparison chart for top funds. |
| P4 | There is limited visibility into investor behaviour patterns such as SIP discontinuation risk. | SIP continuity analysis flags investors whose average instalment gap exceeds 35 days. |

## 4.Technology Stack:

| Category | Tools |
|---|---|
| Language | Python 3 |
| Data Manipulation | Pandas |
| Numerical / Statistics | NumPy, SciPy (linregress) |
| Visualisation | Matplotlib, Seaborn |
| Visualisation (interactive) | Plotly |
| Database | SQLite |
| ORM | SQLAlchemy |
| Notebook Environment | Jupyter Notebook |
| Dashboard | Power BI Desktop |
| Version Control | Git, GitHub |
| Live Data | requests (mfapi.in REST API) |

## 5.Dataset Description:

Ten raw datasets feed the pipeline, integrated into a single SQLite database.

| Dataset | Rows | Description |
|---|---|---|
| fund_master.csv | 40 | Scheme master: AMFI code, fund house, category, expense ratio, risk grade |
| nav_history.csv | 41,720 | Daily NAV per scheme, 2022-2025 |
| scheme_performance.csv | 40 | 1/3/5yr returns, Sharpe, Sortino, Max Drawdown, AUM, expense ratio |
| investor_transactions.csv | 35,144 | SIP / Lumpsum / Redemption transactions for 5,000 investors |
| aum_by_fund_house.csv | 160 | Quarterly AUM by fund house |
| monthly_sip_inflows.csv | 48 | Month-wise industry SIP inflow and active account counts |
| portfolio_holdings.csv | 240 | Top equity holdings by stock, sector, and weight % |
| benchmark_indices.csv | 3,129 | Daily benchmark index closes (NIFTY 100/50 TRI, BSE 250 SmallCap TRI) |
| category_inflows.csv | 240 | Month x category net inflow, FY 2024-25 |
| industry_folio_count.csv | 48 | Month-wise total investor folio count |
| HDFC_Top100.csv | 1,043 | Individual raw NAV file for HDFC Top 100, fetched live from mfapi.in |

# 6.Project Architecture:

The platform follows a standard Extract → Transform → Load → Analyse → Visualise pipeline.


## Layer Details

| Layer | Description |
| --- | --- |
| Extract | Raw CSVs plus a live NAV fetch from api.mfapi.in for one scheme (HDFC Top 100) |
| Transform | Pandas: deduplicate, forward-fill NAV, standardise types, validate ranges |
| Load | SQLAlchemy loads cleaned data into a 7-table SQLite database |
| Analyze | Jupyter notebooks: EDA, CAGR/Sharpe/Sortino/Alpha-Beta, VaR, HHI, cohorts |
| Visualize | Power BI: 4-page dashboard with KPIs, slicers, and drill-through |

## 7.ETL Pipeline:

The ETL pipeline (**scripts/etl_pipeline.py**) loads each cleaned CSV into its target SQLite table using a simple file-to-table mapping and `to_sql(if_exists="replace")`.

| Source File | Target Table |
|---|---|
| clean_fund_master.csv | dim_fund |
| clean_nav.csv | fact_nav |
| clean_transactions.csv | fact_transactions |
| clean_performance.csv | fact_performance |
| clean_portfolio_holdings.csv | fact_portfolio |
| clean_monthly_sip_inflows.csv | fact_sip_industry |
| clean_aum_by_fund_house.csv | fact_aum |

Each notebook (01 and 02) also performs its own ingestion and cleaning independently, so the analysis is fully reproducible directly from the raw files without depending on the script pipeline.

## 8.Data Cleaning & Preprocessing:

Cleaning rules applied per dataset:

- **NAV history**: sorted by fund and date, duplicates removed, NAV validated > 0, daily return computed via percentage change.
- **Investor transactions**: transaction type standardised (e.g. "Sip" → "SIP"), amounts validated > 0, invalid dates dropped.
- **Scheme performance**: Sharpe ratio coerced to numeric, negative-Sharpe funds flagged, expense ratio checked against the 0.1%-2.5% valid range.
- **All other files**: duplicate rows removed, fully-empty rows dropped, text columns filled and stripped, numeric columns coerced where possible.

### Data Quality Summary

| Check | Result |
|---|---|
| fund_master rows | 40 |
| nav rows | 41,720 |
| transactions rows | 35,144 |
| performance rows | 40 |
| NAV nulls | 0 |
| NAV <= 0 rows | 0 |
| Duplicate (amfi_code, date) in nav | 0 |
| Transactions with amount <= 0 | 0 |
| Orphan AMFI codes in nav | 0 |

## 9.Database Design:

The cleaned data is loaded into **bluestock_mf.db**, a SQLite database with a 7-table schema. **fact_performance** is intentionally denormalised — it carries scheme name, fund house, category, AUM, and expense ratio directly, so most analytical queries need no joins.

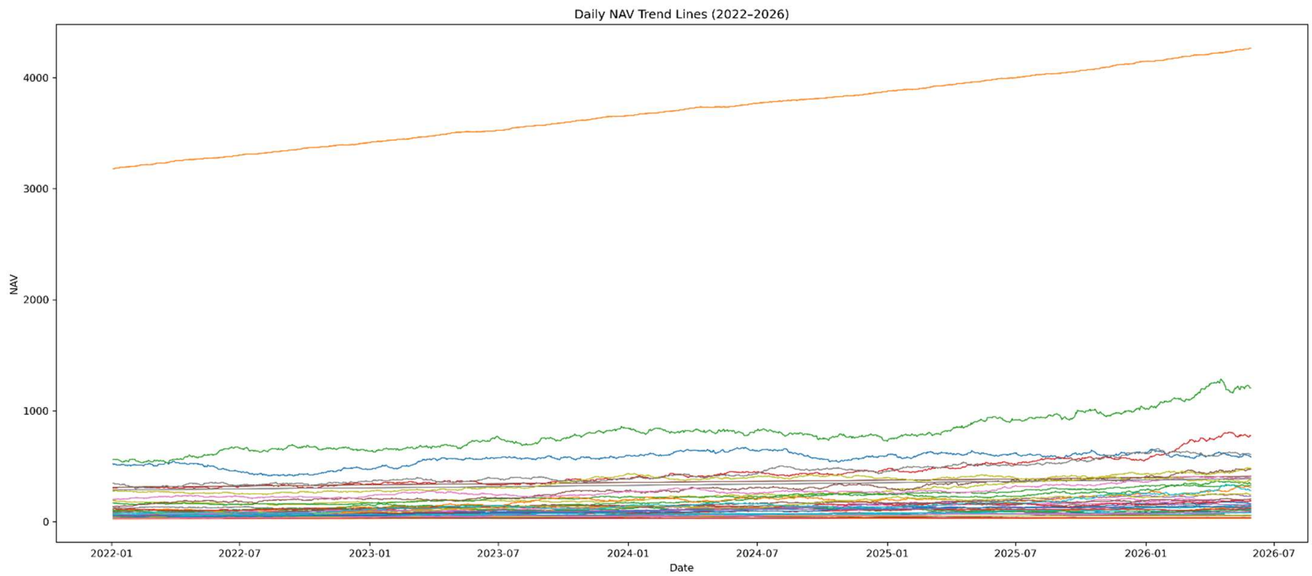
| Table | Type | Key Columns |
|---|---|---|
| dim_fund | Dimension | amfi_code (PK) |
| fact_nav | Fact | amfi_code, date |
| fact_transactions | Fact | investor_id, amfi_code, date |
| fact_performance | Fact (denormalised) | amfi_code, scheme_name, fund_house |
| fact_aum | Fact | fund_house, date |
| fact_portfolio | Fact | amfi_code, stock_symbol |
| fact_sip_industry | Fact | month |

### Sample Analytical Queries

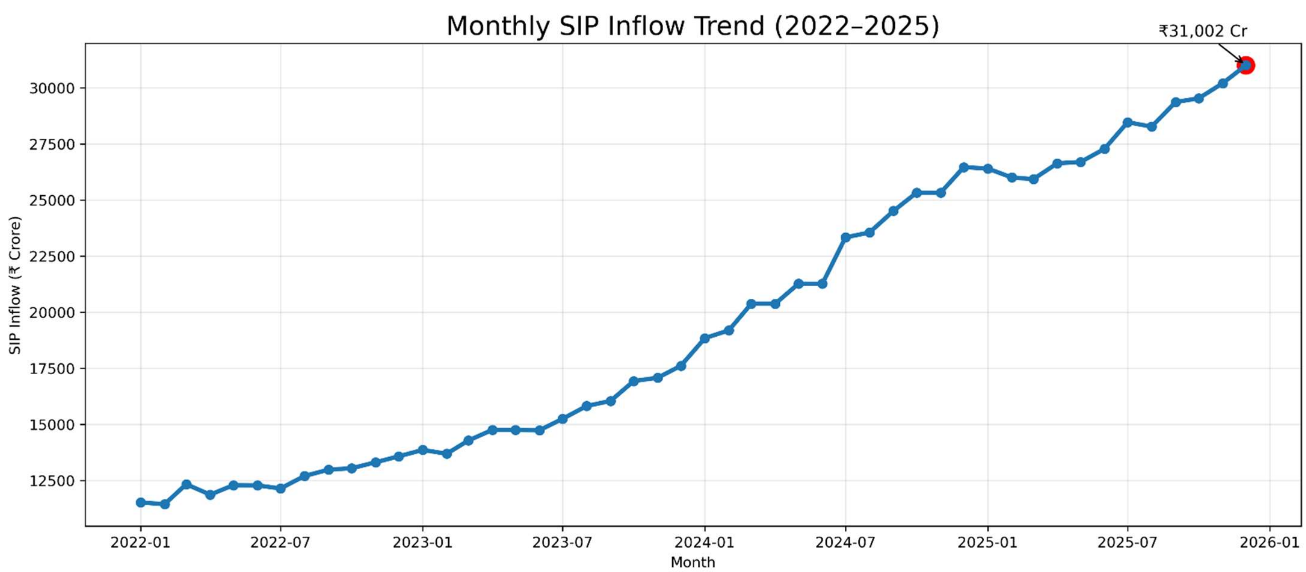
10 flat SQL queries were written against this schema (full list in **sql/queries.sql**), for example:

- Top 5 funds by AUM
- Average NAV per month
- SIP inflow YOY growth
- Transactions by state
- Funds with expensive ratio < 1%
- Total number of transactions
- Fund count by category
- Total SIP transactions
- Average expensive ratio by category
- Maximum nav

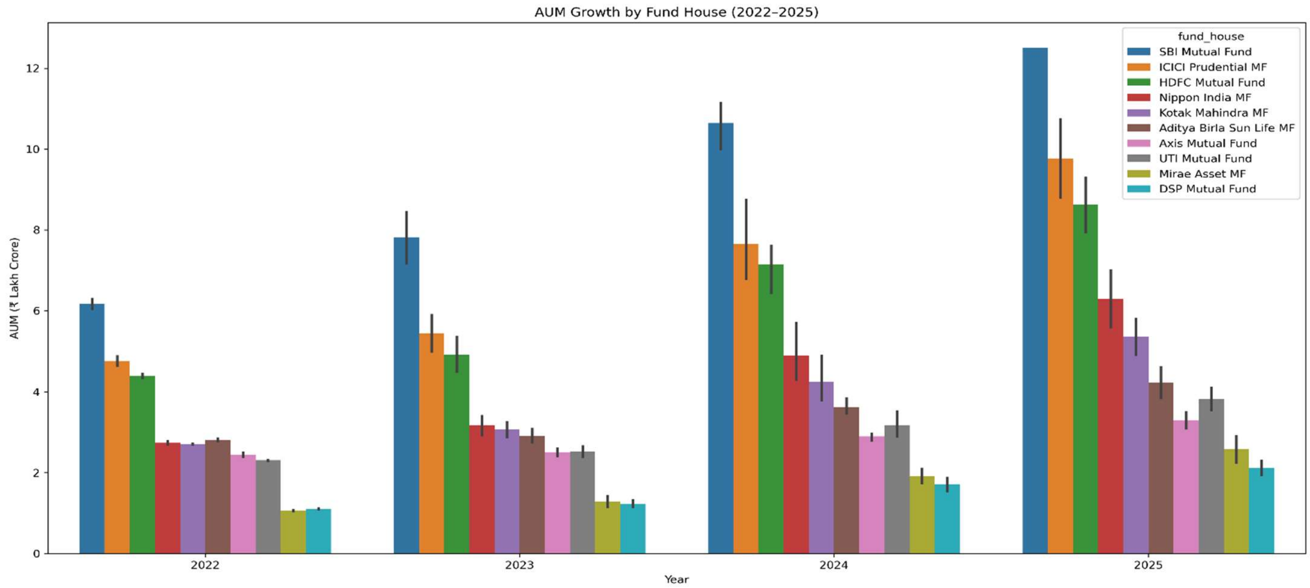
# 10.Exploratory Data Analysis:



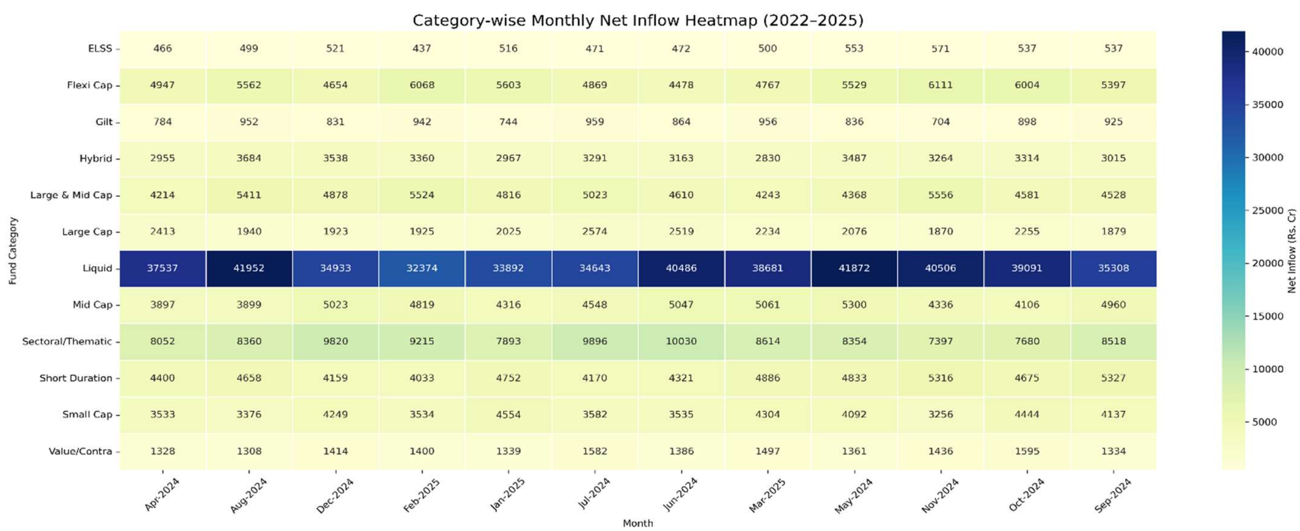

Daily NAV trend across all 40 tracked schemes shows broadly positive growth, with wider dispersion in Mid Cap and Small Cap categories.



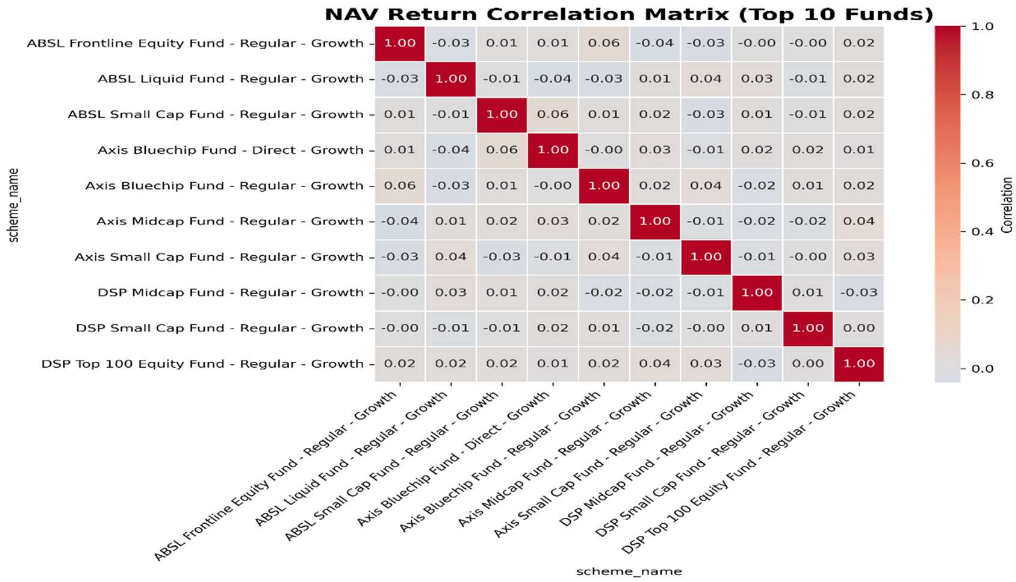

Monthly SIP inflow shows a consistent upward trend across the analysis period.




AUM growth by fund house, 2022-2025.

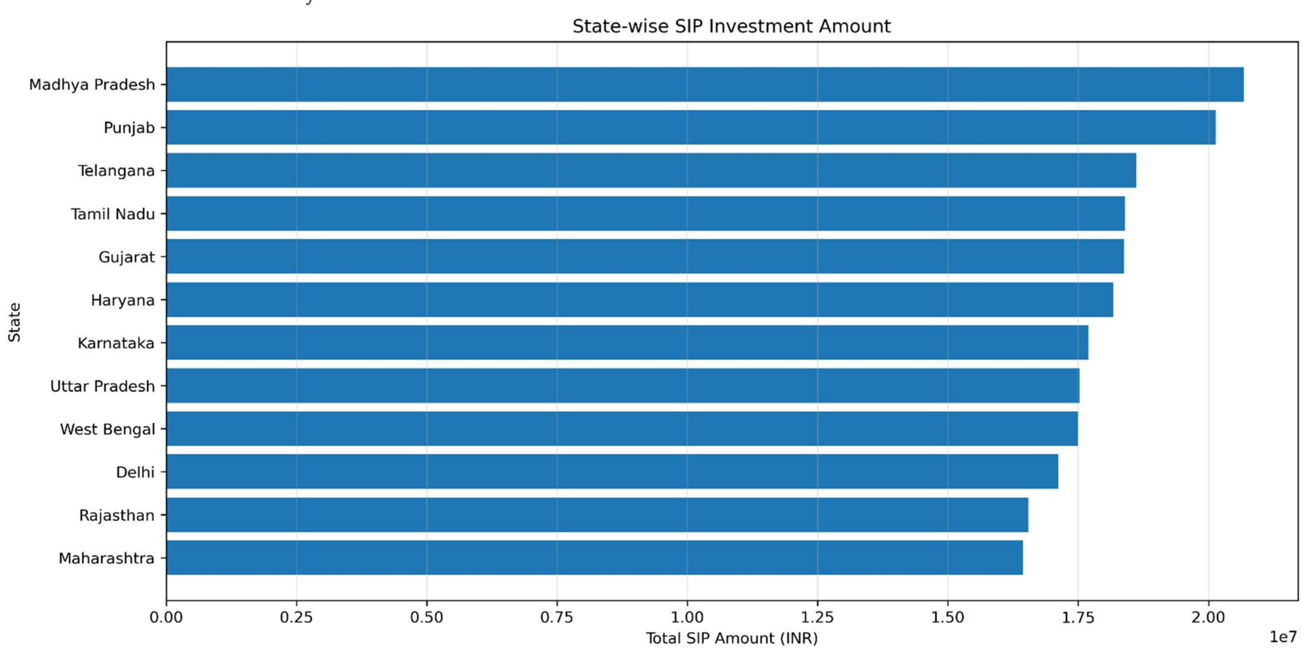


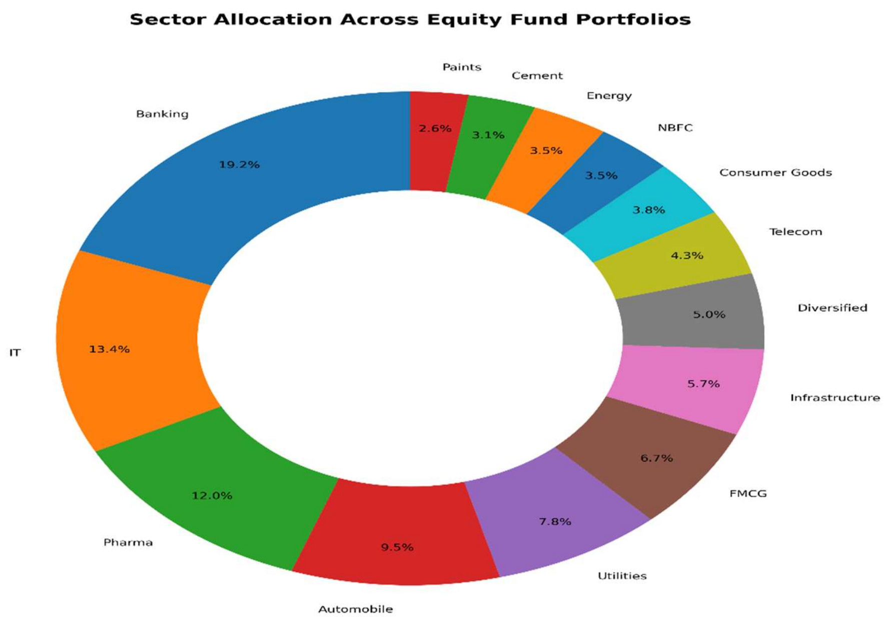

Category-wise net inflow heatmap, FY 2022-25.



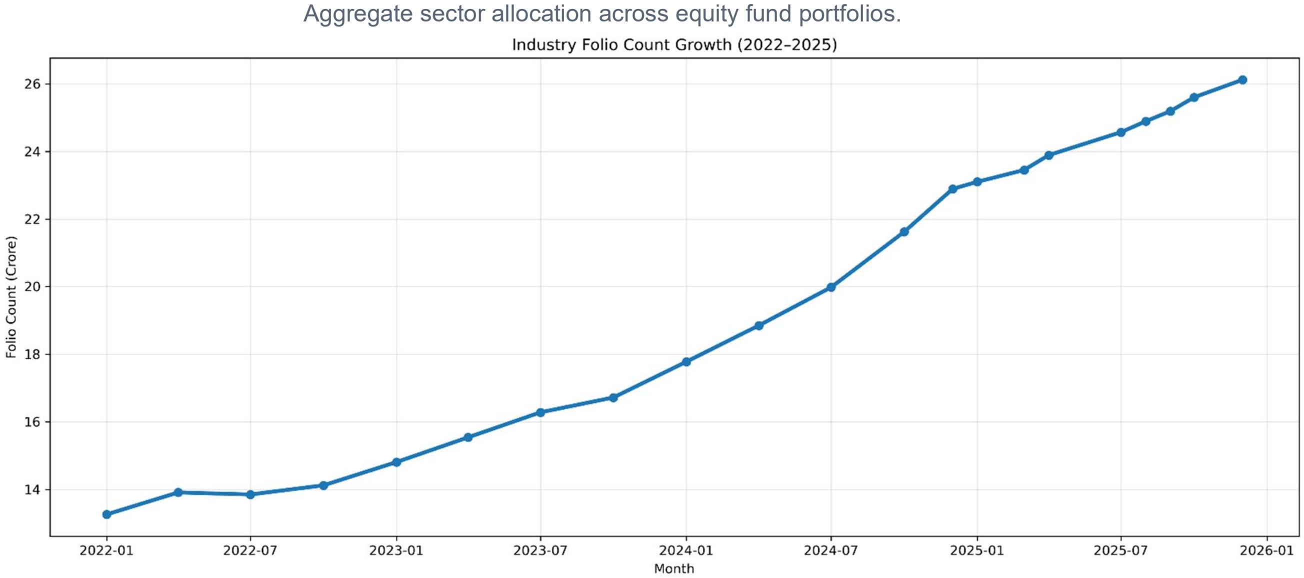

Funds in the same category are more correlated with each other, supporting diversification across categories rather than within one.

Transaction amount by state.






Aggregate sector allocation across equity fund portfolios.

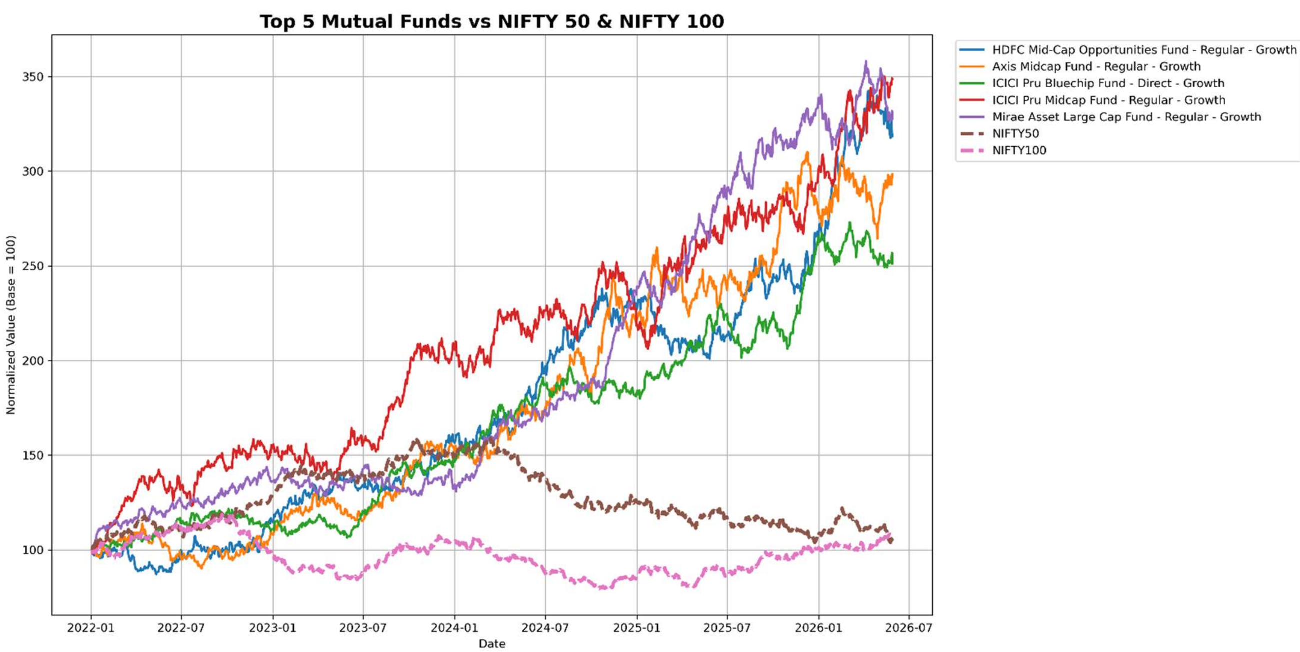



Industry folio count growth, 2022-2026.

## 11.Performance Analytics:

CAGR, Sharpe Ratio, Sortino Ratio, Alpha/Beta (linear regression against NIFTY 100 TRI), and Maximum Drawdown were computed per scheme. A composite Fund Scorecard (0-100) blends these into a single ranking: 30% CAGR rank, 25% Sharpe rank, 20% Alpha rank, 15% expense-ratio rank (inverse), 10% drawdown rank.

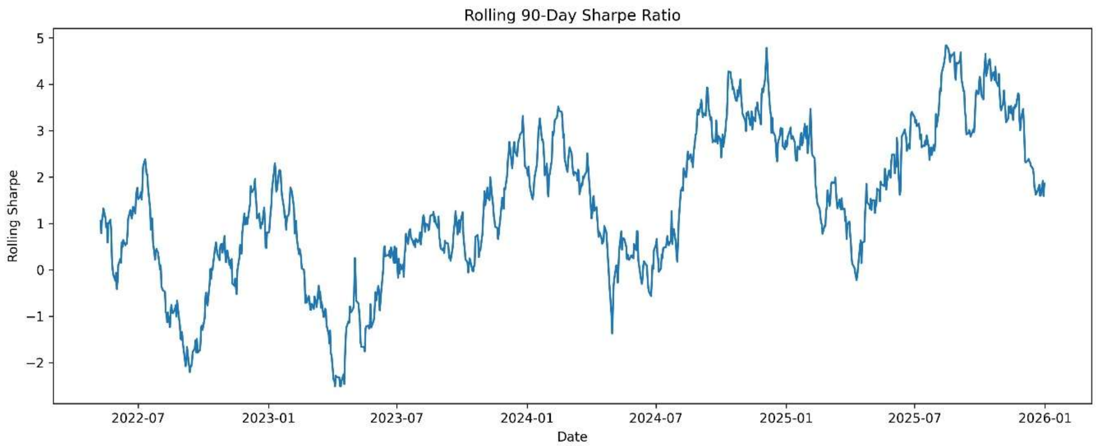
| Scheme | Score | Sharpe | CAGR % | Alpha |
|---|---|---|---|---|
| Axis Large Cap Fund | 98.1 | 1.39 | 32.63 | 27.88 |
| Axis Mid Cap Fund | 88.9 | 1.0 | 24.39 | 23.19 |
| Mirae Asset Mid Cap Fund | 86.2 | 1.07 | 25.6 | 20.7 |
| ICICI Prudential Small Cap Fund | 81.9 | 0.78 | 20.06 | 18.93 |
| SBI Large Cap Fund | 81.5 | 0.97 | 24.84 | 22.29 |




Top 5 scorecard funds vs the NIFTY 100 TRI benchmark, rebased to a common start of 100.

### Risk Metrics:

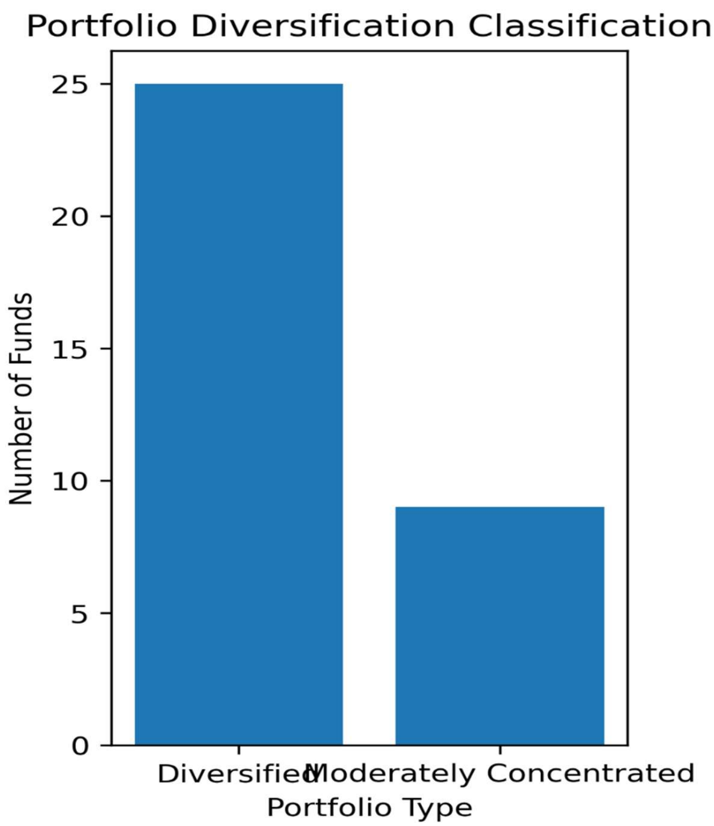
| AMFI Code | VaR (95%, daily) | CVaR (95%, daily) |
|---|---|---|
| 119443 | -1.71% | -2.07% |
| 119406 | -1.69% | -2.14% |
| 119147 | -1.68% | -2.06% |




Rolling 90-day Sharpe Ratio for a representative fund, showing how risk-adjusted performance changes over time rather than staying fixed.

## Sector Concentration (HHI)

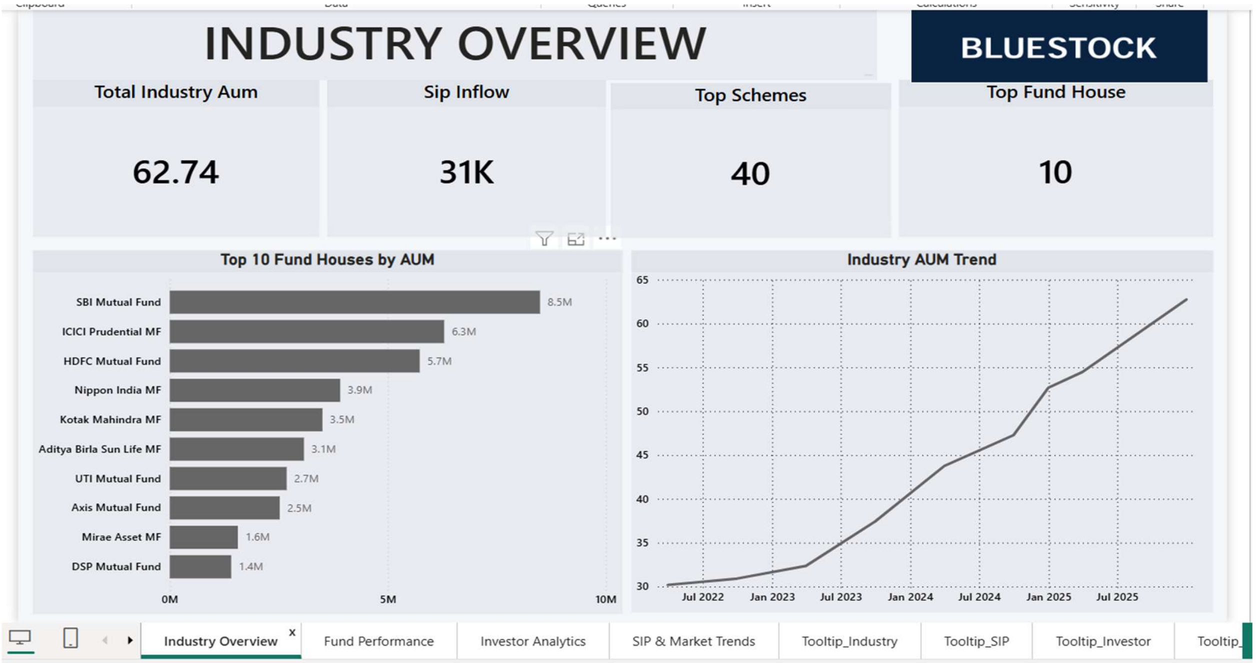
| AMFI Code | Sector HHI | Concentration Risk |
|---|---|---|
| 118555 | 1012.1 | Low |
| 118481 | 937.4 | Low |
| 118518 | 868.0 | Low |



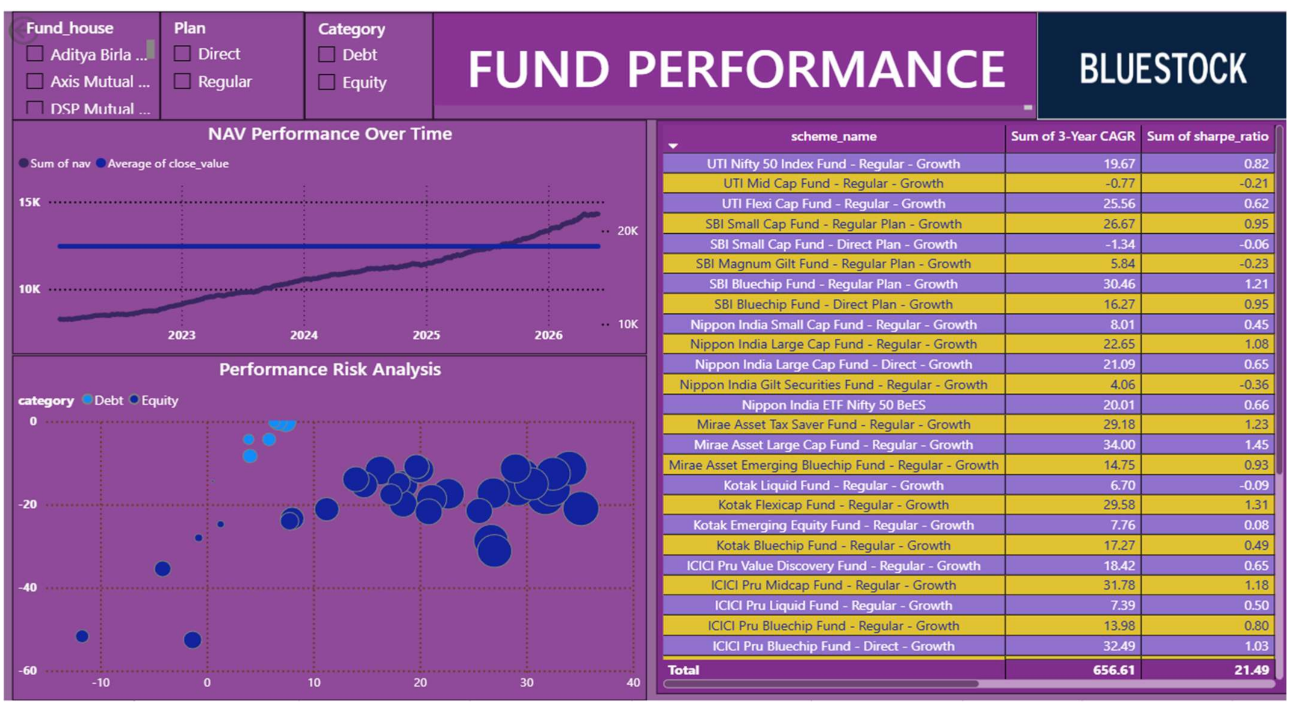

Fund count by sector concentration risk band.

## 12.Power BI Dashboard:

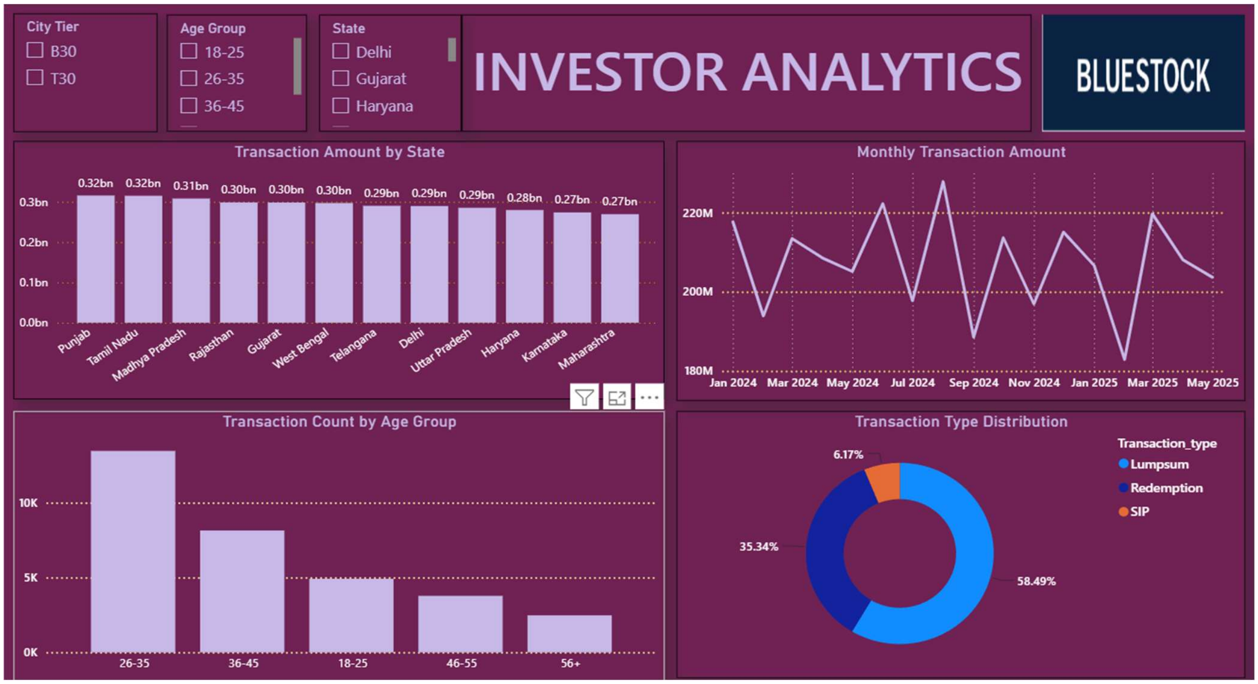
The dashboard has 4 pages — Industry Overview, Fund Performance, Investor Analytics, and SIP & Market Trends — connected to the SQLite database via cleaned CSV imports, with slicers and cross-filtering between visuals.



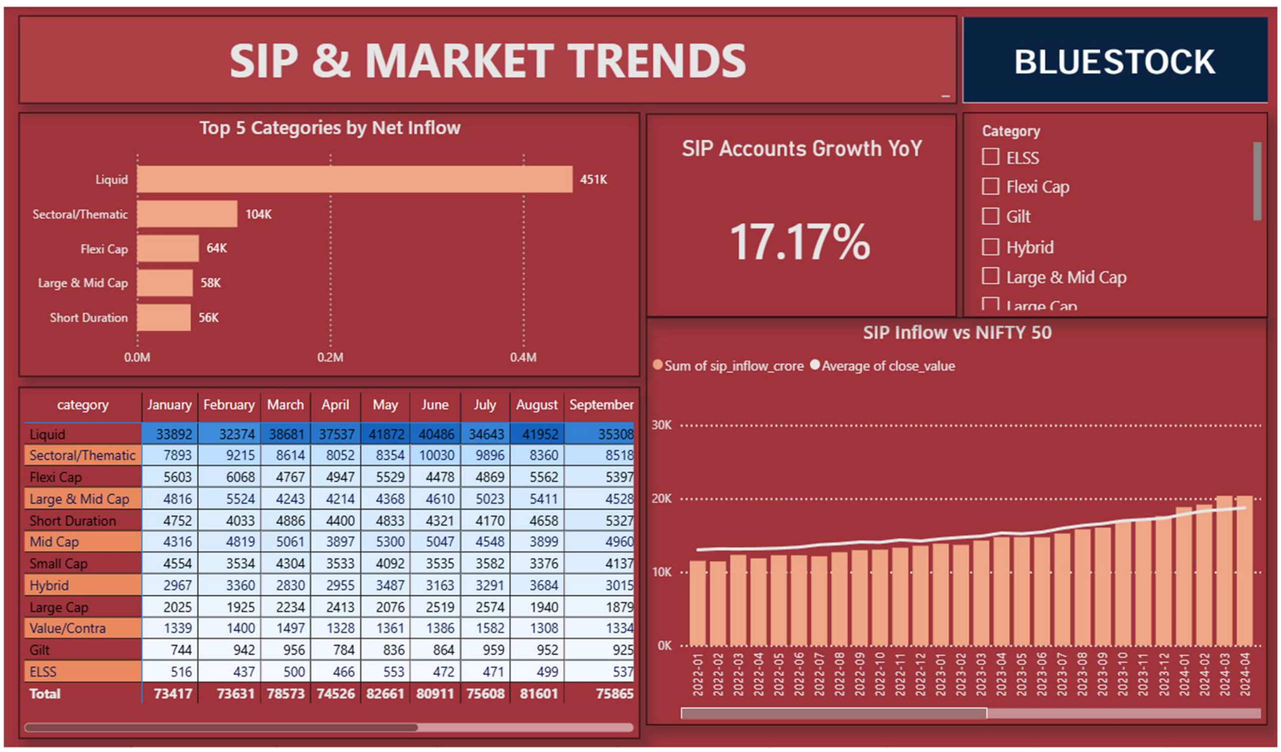

Industry Overview: total industry AUM, SIP inflow, top fund houses by AUM,Top schemes,Top 10 fund houses by AUM, Industry AUM trend.




Fund Performance: scorecard table, return vs risk scatter plot,NAV performance,Fund_House,Plan,Category.




Investor Analytics: transaction amount by state, SIP demographics, Age group,Distribution,City_tier,Age group,State.




| category | January | February | March | April | May | June | July | August | September |
|---|---|---|---|---|---|---|---|---|---|
| Liquid | 33892 | 32374 | 38681 | 37537 | 41872 | 40486 | 34643 | 41952 | 35308 |
| Sectoral/Thematic | 7893 | 9215 | 8614 | 8052 | 8354 | 10030 | 9896 | 8360 | 8518 |
| Flexi Cap | 5603 | 6068 | 4767 | 4947 | 5529 | 4478 | 4869 | 5562 | 5397 |
| Large & Mid Cap | 4816 | 5524 | 4243 | 4214 | 4368 | 4610 | 5023 | 5411 | 4528 |
| Short Duration | 4752 | 4033 | 4886 | 4400 | 4833 | 4321 | 4170 | 4658 | 5327 |
| Mid Cap | 4316 | 4819 | 5061 | 3897 | 5300 | 5047 | 4548 | 3899 | 4960 |
| Small Cap | 4554 | 3534 | 4304 | 3533 | 4092 | 3535 | 3582 | 3376 | 4137 |
| Hybrid | 2967 | 3360 | 2830 | 2955 | 3487 | 3163 | 3291 | 3684 | 3015 |
| Large Cap | 2025 | 1925 | 2234 | 2413 | 2076 | 2519 | 2574 | 1940 | 1879 |
| Value/Contra | 1339 | 1400 | 1497 | 1328 | 1361 | 1386 | 1582 | 1308 | 1334 |
| Gilt | 744 | 942 | 956 | 784 | 836 | 864 | 959 | 952 | 925 |
| ELSS | 516 | 437 | 500 | 466 | 553 | 472 | 471 | 499 | 537 |
| **Total** | **73417** | **73631** | **78573** | **74526** | **82661** | **80911** | **75608** | **81601** | **75865** |

SIP & Market Trends: category inflow table, SIP inflow vs NIFTY 50,Top 5 category By netflow, SIP accounts growth, Category

## 13.Key Findings:

- NAV generally increased across most tracked schemes over the analysis period, though Mid Capand Small Cap schemes showed materially wider dispersion in outcomes.
- The Rolling Sharpe Ratio identifies periods of changing risk-adjusted performance that a singlepoint-in-time Sharpe ratio would miss entirely.
- Cohort analysis shows the 2022 cohort contributes the largest share of both investor count and totalcapital invested, suggesting strong early-investor retention.
- SIP continuity analysis flags 1,609 of 1,610 eligible long-tenure investors (99.9%) with an averageinstalment gap beyond 35 days — a strong signal for retention outreach.
- Sector HHI quantifies concentration risk: a small number of funds are materially more concentratedin a handful of sectors than the rest of the universe.
- Funds within the same category show meaningfully higher return correlation than funds acrosscategories, supporting cross-category diversification.

## 14.Challenges & Solutions:

| Challenge | Solution |
|---|---|
| Power BI relationship between the daily Dim_Date table and the monthly SIP inflow table had mismatched grain, causing the benchmark comparison line to show a flat, incorrect average. | Rebuilt the relationship on a shared Year-Month key with correct 1:Many cardinality (monthly table = 1, Dim_Date = Many) and Both-direction cross-filtering. |
| The date field in the monthly SIP/category tables was stored as text ("2025-12"), which cannot directly join against a real date column. | Added a Year-Month text column to Dim_Date and to each monthly table so they can be joined directly on matching text keys. |
| Sort order on Month and Weekday columns defaulted to alphabetical rather than chronological, breaking the x-axis order on several dashboard charts. | Added explicit "sort by column" rules (Month by Month Number, Weekday by Weekday Number) so charts render in true chronological order. |
| The live NAV fetch from api.mfapi.in requires outbound internet access, which is not always available in a sandboxed analysis environment. | Wrapped the API call in a try/except that falls back to a locally cached NAV snapshot, so the notebook still runs end-to-end offline while preserving the original live-fetch code. |
| scheme_performance.csv needed to carry expense_ratio_pct directly, since several downstream cells reference it without joining back to fund_master. | Adjusted the raw data schema so expense_ratio_pct is generated directly into scheme_performance.csv, matching what the downstream notebooks expect. |

## 15. Conclusion:

This project demonstrates a complete analytics workflow from raw financial data to executive dashboards and risk analytics — covering data engineering (ETL, database design), quantitative analysis (performance and risk metrics), and business intelligence (interactive dashboarding). The resulting fund scorecard, risk flags, and investor-behaviour insights give a concrete, reproducible basis for the kind of decisions a retail investor or advisor actually needs to make.

## 16.Future Enhancements:

- Integrate live AMFI / mfapi.in APIs across all 40 tracked schemes, not just one.
- Replace the rule-based fund recommendation score with a machine-learning model trained onhistorical risk-adjusted outcomes.
- Deploy the Power BI dashboard to Power BI Service for stakeholder self-service access.
- Automate the ETL pipeline on a scheduled run (e.g. daily NAV refresh).
- Add portfolio-level analysis so an investor can evaluate a basket of funds together, not just onescheme at a time.

## 17. References:


- Association of Mutual Funds in India (AMFI) — www.amfiindia.com
- mfapi.in — public REST API for historical and live mutual fund NAV data
- SEBI Mutual Fund Regulations — risk categorisation and disclosure norms
- Pandas, NumPy, SciPy, Matplotlib, Seaborn, SQLAlchemy — official project documentation
- Power BI documentation — Microsoft Learn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